

Farm Exposure: Ovigerous female sea lice

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Individual sea lice dataset (SL_Indiv)

```
SL_Indiv<-read.table("IndividualFishDataset.csv",  
  header = TRUE,sep = ',',fill = TRUE,dec = ".",na.strings = "NA")
```

```
library(ggplot2)  
library(glmmTMB)  
suppressMessages(library(lmtest))  
library(performance)  
library(gridExtra)  
library(grid)
```

Data adjustment before analysis

The outliers were identified: sites with less than 60 inspections, fish that stayed more than 24 months in the seawater, outliers (sea lice counts higher than 161). In addition, data from full production cycles of fish were kept.

```
# Excluding fish stock transported to sea water in 2023  
SL_Indiv1<-SL_Indiv[!(SL_Indiv$Year.class %in% c("Atlantic Salmon 2023 S1/2",  
  "Atlantic Salmon 2023")),]  
  
# Sites with less than 60 inspections  
Insp<-aggregate(SL_Indiv1$SampleID,list(SL_Indiv1$SiteName),  
  function(x){length(unique(x))})  
Insp_Sel<-Insp[Insp$x<60,]$Group.1  
  
# Fish more than 24 months in the sea water, group '30' includes 25-30 in seawater  
SL_Indiv2<-SL_Indiv1[SL_Indiv1$TimeSea<30,]  
SL_Indiv3<-SL_Indiv2[!(SL_Indiv2$SiteName %in% Insp_Sel),]  
  
# Exclude observation with 7 farms per bay due to low number of data points  
table(SL_Indiv3$Active_Farm)
```

```
##  
##      1      2      3      4      5      6      7  
## 78204 67577 11821 7263 1374 1300 180
```

```
unique(SL_Indiv3[SL_Indiv3$Active_Farm==7,]$SiteName)
```

```
## [1] "Daonish" "Golam" "Ardmore"
```

```
SL_Indiv4<-SL_Indiv3[SL_Indiv3$Active_Farm<7,]
```

```
# Outliers
```

```
SL_Fish<-SL_Indiv4[SL_Indiv4$LepeophtheirusSalmonisEggs<60,]
```

Comparison of raw and cleaned datasets

```
# Number of counts per fish
```

```
nrow(SL_Indiv)
```

```
## [1] 186796
```

```
nrow(SL_Fish)
```

```
## [1] 167537
```

```
# Number of salmon farms
```

```
length(unique(SL_Indiv$SiteName))
```

```
## [1] 41
```

```
length(unique(SL_Fish$SiteName))
```

```
## [1] 27
```

Exploration of proportion between total mobile and total adult female life stages

```
# Overall proportion of Total adult females and Total mobiles
```

```
sum(SL_Fish$LepeophtheirusSalmonisEggs)/sum(SL_Fish$LepeophtheirusSalmonisTotal)
```

```
## [1] 0.1762836
```

```
# Difference in proportion of months
```

```
Eggs<-tapply(SL_Fish$LepeophtheirusSalmonisEggs,SL_Fish$NewMonth,mean)
```

```
Mobile<-tapply(SL_Fish$LepeophtheirusSalmonisTotal,SL_Fish$NewMonth,mean)
```

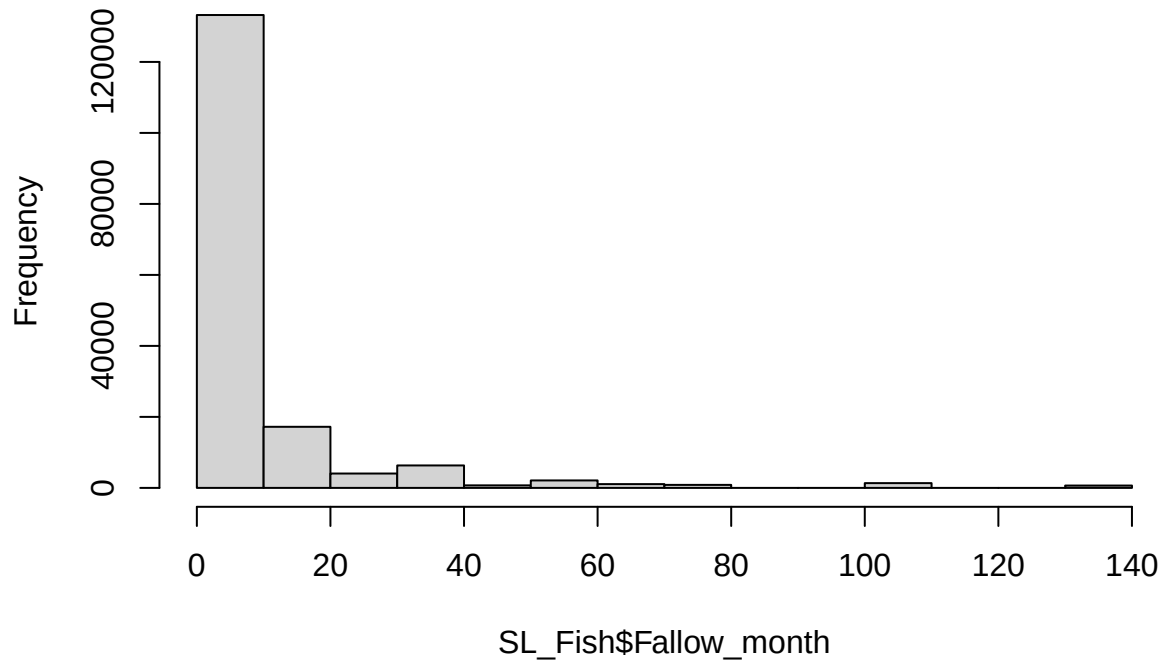
```
Eggs/Mobile
```

```
##      Apr1      Apr2      Aug      Feb  Jan/Dec      Jul      Jun      Mar1  
## 0.1011465 0.1143907 0.2243866 0.1333056 0.1698432 0.2554668 0.2625916 0.1202328  
##      Mar2      May1      May2      Nov      Oct      Sep  
## 0.1258729 0.1014337 0.1261846 0.1880703 0.2233135 0.2485315
```

Adjustment of the format of Fallow period

```
# Values higher than 25 months were substituted by 25
hist(SL_Fish$Fallow_month)
```

Histogram of SL_Fish\$Fallow_month



```
SL_Fish[SL_Fish$Fallow_month>24,]$Fallow_month<-25
table(SL_Fish$Fallow_month)
```

```
##
##      0      1      2      3      4      5      6      7      8      9     10     11     12
## 20310 1310  8152  9840 13517 11271 13982 15892 14554 14389 10003  3237   712
##    13    14    15    16    17    18    19    20    21    25
##  2526  1660  1083  2929  1117  1138  2304   503   665 16443
```

The format of variables in the dataset (SL_Fish), sea lice values recorded as counts per fish, was adjusted.

```
SL_Fish$Year_f<-as.factor(SL_Fish$Year)
SL_Fish$TimeSea_f<-as.factor(SL_Fish$TimeSea)
SL_Fish$Year_n<-as.numeric(SL_Fish$Year_f)/10
SL_Fish$Month_n<-as.numeric(SL_Fish$Month)/10
```

A new dataset (SL_Pen), sea lice values recorded as a mean value per fish per pen, was created.

```
SL_Pen<-aggregate(SL_Fish$LepeophtheirusSalmonisEggs,list(
  SL_Fish$Year_n,SL_Fish$Year,SL_Fish$SiteName,SL_Fish$Month,SL_Fish$Month_n,
  SL_Fish$NewMonth,SL_Fish$S.R,SL_Fish$Class,SL_Fish$TimeSea,SL_Fish$Date,
  SL_Fish$New_Comp,SL_Fish$SampleID,SL_Fish$Longitude,SL_Fish$Latitude,
  SL_Fish$Temp,SL_Fish$Sal,SL_Fish$Oxy1,SL_Fish$Depth_FINAL,SL_Fish$Distance_Mouth,
  SL_Fish$Length,SL_Fish$Width_Mouth,SL_Fish$Fallow_month,SL_Fish$Active_Farm),mean)
colnames(SL_Pen)<-c('Year','Year_f','SiteName','Month','Month_n','NewMonth',
  'S.R','Class','TimeSea','Date','Company','ID',
  'Longitude','Latitude','Temp','Sal','Oxy','Depth',
  'Distance_Mouth','Length','Width_Mouth','Fallow',
  'Active_Farms','Lep')
```

The format of variables in the dataset (SL_Pen) was adjusted.

```
SL_Pen$TimeSea<-as.factor(SL_Pen$TimeSea)

# Number of mean values per farm in the dataset
nrow(SL_Pen)
```

```
## [1] 5723
```

A new dataset (SL_Farm), sea lice values recorded as a mean value per fish per farm, was created.

```
SL_Farm<-aggregate(SL_Fish$LepeophtheirusSalmonisEggs,list(
  SL_Fish$Year_n,SL_Fish$Year,SL_Fish$SiteName,SL_Fish$Month,SL_Fish$Month_n,
  SL_Fish$NewMonth,SL_Fish$Class,SL_Fish$TimeSea,SL_Fish$Date,
  SL_Fish$New_Comp,SL_Fish$Depth_FINAL,SL_Fish$Distance_Mouth,
  SL_Fish$Length,SL_Fish$Width_Mouth,SL_Fish$Longitude,SL_Fish$Latitude,
  SL_Fish$Temp,SL_Fish$Sal,SL_Fish$Oxy1,SL_Fish$Date,SL_Fish$Fallow_month,
  SL_Fish$Active_Farm),mean)
colnames(SL_Farm)<-c('Year','Year_f','SiteName','Month','Month_n','NewMonth',
  'Class','TimeSea','Date','Company','Depth',
  'Distance_Mouth','Length','Width_Mouth','Longitude',
  'Latitude','Temp','Sal','Oxy','Date','Fallow',
  'Active_Farms','Lep')
```

The format of variables in the dataset (SL_Farm) was adjusted.

```
SL_Farm$TimeSea<-as.factor(SL_Farm$TimeSea)

# Number of mean values per farm in the dataset
nrow(SL_Farm)
```

```
## [1] 2993
```

```
mean(SL_Farm$Lep);sd(SL_Farm$Lep)
```

```
## [1] 0.5904071
```

```
## [1] 1.74786
```

The temporal autocorrelation for models at the fish level (count of sea lice per fish), pen level (mean value of sea lice per fish per pen) and at the farm (mean value of sea lice per fish per farm) level were compared.

Fish level

The SL_Fish dataset (count per fish) was used in a GLM that included all relevant predictors.

```
m1<-glmmTMB(LepeophtheirusSalmonisEggs~scale(Depth_FINAL)+
  scale(Distance_Mouth)+scale(Length)+scale(Width_Mouth)+scale(Active_Farms)+
  scale(Temp)+scale(Sal)+scale(Oxy1)+Year_n+Month_n+TimeSea_f+
  Class+New_Comp+S.R+scale(Fallow_month)+scale(Longitude)+scale(Latitude),
  family=nbinom2,data=SL_Fish)
```

The model was checked for overdispersion, zeroinflation and temporal autocorrelation.

```
# Overdispersion
E2 <- resid(m1, type = "pearson")
N <- nrow(SL_Fish)
p <- length(coef(m1))
sum(E2^2) / (N - p)
```

```
## [1] 1.542729
```

```
# Zero-inflation
check_zeroinflation(m1)
```

```
## # Check for zero-inflation
##
##   Observed zeros: 137265
##   Predicted zeros: 137039
##           Ratio: 1.00
```

```
## Model seems ok, ratio of observed and predicted zeros is within the
##   tolerance range (p = 0.104).
```

```
# Temporal autocorrelation
dwtest(m1)
```

```
##
##   Durbin-Watson test
##
## data:  m1
## DW = 1.0789, p-value < 2.2e-16
## alternative hypothesis: true autocorrelation is greater than 0
```

```
bgtest(m1)
```

```
##
##   Breusch-Godfrey test for serial correlation of order up to 1
##
## data:  m1
## LM test = 35600, df = 1, p-value < 2.2e-16
```

Pen level

The SL_Pen dataset (mean value per fish per pen) was used in a GLM that included all relevant predictors.

```
m2<-glmmTMB(Lep~scale(Depth)+scale(Distance_Mouth)+scale(Length)+
  scale(Width_Mouth)+scale(Active_Farms)+scale(Temp)+scale(Sal)+
  scale(Oxy)+Year+Month_n+TimeSea+Class+Company+S.R+scale(Fallow)+
  scale(Longitude)+scale(Latitude),
  ziformula=~scale(Depth)+scale(Distance_Mouth)+scale(Length)+
  scale(Width_Mouth)+scale(Active_Farms)+scale(Temp)+scale(Sal)+
  scale(Oxy)+Year+Month_n+TimeSea+Class+Company+-S.R+scale(Fallow)+
  scale(Longitude)+scale(Latitude),
  family=ziGamma(link="log"),data=SL_Pen)
```

The temporal autocorrelation was tested.

```
dwtest(m2)
```

```
##
## Durbin-Watson test
##
## data: m2
## DW = 1.1625, p-value < 2.2e-16
## alternative hypothesis: true autocorrelation is greater than 0
```

```
bgtest(m2)
```

```
##
## Breusch-Godfrey test for serial correlation of order up to 1
##
## data: m2
## LM test = 1005.5, df = 1, p-value < 2.2e-16
```

Farm level

The SL_Farm dataset (mean value per fish per farm) was used in a GLM that included all relevant predictors.

```
m3<-glmmTMB(Lep~scale(Depth)+scale(Distance_Mouth)+scale(Length)+
  scale(Width_Mouth)+scale(Active_Farms)+scale(Temp)+scale(Sal)+
  scale(Oxy)+Year+Month_n+TimeSea+Class+Company+scale(Fallow)+
  scale(Longitude)+scale(Latitude),
  ziformula=~scale(Depth)+scale(Distance_Mouth)+scale(Length)+
  scale(Width_Mouth)+scale(Active_Farms)+scale(Temp)+scale(Sal)+
  scale(Oxy)+Year+Month_n+TimeSea+Class+Company+scale(Fallow)+
  scale(Longitude)+scale(Latitude),
  family=ziGamma(link="log"),data=SL_Farm)
```

The model was then tested for temporal autocorrelation and the results were compared to the model (m1) at the fish level and m2 at the pen level.

```
dwtest(m3)
```

```
##
## Durbin-Watson test
##
## data: m3
## DW = 1.4038, p-value < 2.2e-16
## alternative hypothesis: true autocorrelation is greater than 0
```

```
bgtest(m3)
```

```
##
## Breusch-Godfrey test for serial correlation of order up to 1
##
## data: m3
## LM test = 268.91, df = 1, p-value < 2.2e-16
```

Lower temporal autocorrelation was detected for the model at the farm level. Based on statistical diagnostics, data availability, and the research question the farm level data (SL_Farm) were used for further analysis.

The model was checked for multicollinearity.

```
check_collinearity(m3)
```

```
## # Check for Multicollinearity
##
## * conditional component:
##
## Low Correlation
##
##      Term  VIF  VIF 95% CI adj. VIF Tolerance Tolerance 95% CI
##      scale(Depth) 1.70 [1.62, 1.79] 1.31 0.59 [0.56, 0.62]
##      scale(Distance_Mouth) 1.99 [1.89, 2.10] 1.41 0.50 [0.48, 0.53]
##      scale(Width_Mouth) 2.19 [2.07, 2.31] 1.48 0.46 [0.43, 0.48]
##      scale(Active_Farms) 1.52 [1.45, 1.59] 1.23 0.66 [0.63, 0.69]
##      scale(Temp) 2.23 [2.12, 2.36] 1.49 0.45 [0.42, 0.47]
##      scale(Sal) 1.81 [1.72, 1.91] 1.34 0.55 [0.52, 0.58]
##      scale(Oxy) 1.20 [1.15, 1.25] 1.09 0.84 [0.80, 0.87]
##      Year 1.69 [1.61, 1.78] 1.30 0.59 [0.56, 0.62]
##      Month_n 2.35 [2.22, 2.48] 1.53 0.43 [0.40, 0.45]
##      TimeSea 1.37 [1.32, 1.44] 1.05 0.73 [0.69, 0.76]
##      Class 1.47 [1.41, 1.55] 1.21 0.68 [0.65, 0.71]
##      Company 3.72 [3.50, 3.95] 1.93 0.27 [0.25, 0.29]
##      scale(Fallow) 1.30 [1.25, 1.36] 1.14 0.77 [0.74, 0.80]
##
## Moderate Correlation
##
##      Term  VIF  VIF 95% CI adj. VIF Tolerance Tolerance 95% CI
##      scale(Length) 5.48 [5.14, 5.84] 2.34 0.18 [0.17, 0.19]
##      scale(Longitude) 6.86 [6.43, 7.33] 2.62 0.15 [0.14, 0.16]
##      scale(Latitude) 8.76 [8.20, 9.37] 2.96 0.11 [0.11, 0.12]
##
```

```
## * zero inflated component:
##
## Low Correlation
##
##          Term   VIF      VIF 95% CI adj. VIF Tolerance Tolerance 95% CI
##          scale(Depth) 1.77 [ 1.69, 1.87]    1.33    0.56    [0.53, 0.59]
##    scale(Distance_Mouth) 2.15 [ 2.04, 2.27]    1.46    0.47    [0.44, 0.49]
##          scale(Width_Mouth) 2.18 [ 2.06, 2.30]    1.47    0.46    [0.43, 0.48]
##          scale(Active_Farms) 1.44 [ 1.38, 1.51]    1.20    0.69    [0.66, 0.73]
##          scale(Temp) 2.01 [ 1.91, 2.12]    1.42    0.50    [0.47, 0.52]
##          scale(Sal) 1.52 [ 1.46, 1.60]    1.23    0.66    [0.62, 0.69]
##          scale(Oxy) 1.16 [ 1.12, 1.22]    1.08    0.86    [0.82, 0.89]
##          Year 1.37 [ 1.31, 1.43]    1.17    0.73    [0.70, 0.76]
##          Month_n 2.14 [ 2.03, 2.26]    1.46    0.47    [0.44, 0.49]
##          TimeSea 1.50 [ 1.43, 1.57]    1.07    0.67    [0.64, 0.70]
##          scale(Fallow) 1.15 [ 1.11, 1.20]    1.07    0.87    [0.83, 0.90]
##
## Moderate Correlation
##
##          Term   VIF      VIF 95% CI adj. VIF Tolerance Tolerance 95% CI
##          scale(Length) 6.21 [ 5.83, 6.63]    2.49    0.16    [0.15, 0.17]
##    scale(Longitude) 5.93 [ 5.56, 6.33]    2.43    0.17    [0.16, 0.18]
##
## High Correlation
##
##          Term   VIF      VIF 95% CI adj. VIF Tolerance Tolerance 95% CI
##    scale(Latitude) 12.23 [11.43, 13.09]    3.50    0.08    [0.08, 0.09]
```

High multicollinearity was detected between length of bay, longitude and latitude and was addressed by removing Longitude as a predictor.

```
m3.1<-glmmTMB(Lep~scale(Depth)+scale(Distance_Mouth)+scale(Length)+
  scale(Width_Mouth)+scale(Active_Farms)+scale(Temp)+scale(Sal)+
  scale(Oxy)+Year+Month_n+TimeSea+Class+Company+scale(Fallow)+
  scale(Latitude),
  ziformula=~scale(Depth)+scale(Distance_Mouth)+scale(Length)+
  scale(Width_Mouth)+scale(Active_Farms)+scale(Temp)+scale(Sal)+
  scale(Oxy)+Year+Month_n+TimeSea+Class+Company+scale(Fallow)+
  scale(Latitude),family=ziGamma(link="log"),data=SL_Farm)
```

The predictors were selected using a backwards step-wise approach.

```
m3.01<-glmmTMB(Lep~scale(Distance_Mouth)+scale(Length)+
  scale(Width_Mouth)+scale(Active_Farms)+scale(Temp)+scale(Sal)+
  scale(Oxy)+Year+Month_n+TimeSea+Class+Company+scale(Fallow)+
  scale(Latitude),
  ziformula=~scale(Distance_Mouth)+scale(Length)+
  scale(Width_Mouth)+scale(Active_Farms)+scale(Temp)+scale(Sal)+
  scale(Oxy)+Year+Month_n+TimeSea+Class+Company+scale(Fallow)+
  scale(Latitude),family=ziGamma(link="log"),data=SL_Farm)
m3.02<-glmmTMB(Lep~scale(Depth)+scale(Length)+
  scale(Width_Mouth)+scale(Active_Farms)+scale(Temp)+scale(Sal)+
  scale(Oxy)+Month_n+Year+TimeSea+Class+Company+scale(Fallow)+
```

```

scale(Latitude),
ziformula=~scale(Depth)+scale(Length)+
scale(Width_Mouth)+scale(Active_Farms)+scale(Temp)+scale(Sal)+
scale(Oxy)+Month_n+Year+TimeSea+Class+Company+scale(Fallow)+
scale(Latitude),family=ziGamma(link="log"),data=SL_Farm)
m3.03<-glmmTMB(Lep~scale(Depth)+scale(Distance_Mouth)+
scale(Width_Mouth)+scale(Active_Farms)+scale(Temp)+scale(Sal)+
scale(Oxy)+Month_n+Year+TimeSea+Class+Company+scale(Fallow)+
scale(Latitude),
ziformula=~scale(Depth)+scale(Distance_Mouth)+
scale(Width_Mouth)+scale(Active_Farms)+scale(Temp)+scale(Sal)+
scale(Oxy)+Month_n+Year+TimeSea+Class+Company+scale(Fallow)+
scale(Latitude),family=ziGamma(link="log"),data=SL_Farm)
m3.04<-glmmTMB(Lep~scale(Depth)+scale(Distance_Mouth)+scale(Length)+
scale(Active_Farms)+scale(Temp)+scale(Sal)+
scale(Oxy)+Month_n+Year+TimeSea+Class+Company+scale(Fallow)+
scale(Latitude),
ziformula=~scale(Depth)+scale(Distance_Mouth)+scale(Length)+
scale(Active_Farms)+scale(Temp)+scale(Sal)+
scale(Oxy)+Month_n+Year+TimeSea+Class+Company+scale(Fallow)+
scale(Latitude),family=ziGamma(link="log"),data=SL_Farm)
m3.05<-glmmTMB(Lep~scale(Depth)+scale(Distance_Mouth)+scale(Length)+
scale(Width_Mouth)+scale(Temp)+scale(Sal)+
scale(Oxy)+Month_n+Year+TimeSea+Class+Company+scale(Fallow)+
scale(Latitude),
ziformula=~scale(Depth)+scale(Distance_Mouth)+scale(Length)+
scale(Width_Mouth)+scale(Temp)+scale(Sal)+
scale(Oxy)+Month_n+Year+TimeSea+Class+Company+scale(Fallow)+
scale(Latitude),family=ziGamma(link="log"),data=SL_Farm)
m3.06<-glmmTMB(Lep~scale(Depth)+scale(Distance_Mouth)+scale(Length)+
scale(Width_Mouth)+scale(Active_Farms)+scale(Sal)+
scale(Oxy)+Month_n+Year+TimeSea+Class+Company+scale(Fallow)+
scale(Latitude),
ziformula=~scale(Depth)+scale(Distance_Mouth)+scale(Length)+
scale(Width_Mouth)+scale(Active_Farms)+scale(Sal)+
scale(Oxy)+Month_n+Year+TimeSea+Class+Company+scale(Fallow)+
scale(Latitude),family=ziGamma(link="log"),data=SL_Farm)
m3.07<-glmmTMB(Lep~scale(Depth)+scale(Distance_Mouth)+scale(Length)+
scale(Width_Mouth)+scale(Active_Farms)+scale(Temp)+
scale(Oxy)+Year+Month_n+TimeSea+Class+Company+scale(Fallow)+
scale(Latitude),
ziformula=~scale(Depth)+scale(Distance_Mouth)+scale(Length)+
scale(Width_Mouth)+scale(Active_Farms)+scale(Temp)+
scale(Oxy)+Year+Month_n+TimeSea+Class+Company+scale(Fallow)+
scale(Latitude),family=ziGamma(link="log"),data=SL_Farm)
m3.08<-glmmTMB(Lep~scale(Depth)+scale(Distance_Mouth)+scale(Length)+
scale(Width_Mouth)+scale(Active_Farms)+scale(Temp)+scale(Sal)+
Month_n+Year+TimeSea+Class+Company+scale(Fallow)+
scale(Latitude),
ziformula=~scale(Depth)+scale(Distance_Mouth)+scale(Length)+
scale(Width_Mouth)+scale(Active_Farms)+scale(Temp)+scale(Sal)+
Month_n+Year+TimeSea+Class+Company+scale(Fallow)+
scale(Latitude),family=ziGamma(link="log"),data=SL_Farm)

```

```

m3.09<-glmmTMB(Lep~scale(Depth)+scale(Distance_Mouth)+scale(Length)+
scale(Width_Mouth)+scale(Active_Farms)+scale(Temp)+scale(Sal)+
scale(Oxy)+Year+TimeSea+Class+Company+scale(Fallow)+
scale(Latitude),
ziformula=~scale(Depth)+scale(Distance_Mouth)+scale(Length)+
scale(Width_Mouth)+scale(Active_Farms)+scale(Temp)+scale(Sal)+
scale(Oxy)+Year+TimeSea+Class+Company+scale(Fallow)+
scale(Latitude),family=ziGamma(link="log"),data=SL_Farm)
m3.10<-glmmTMB(Lep~scale(Depth)+scale(Distance_Mouth)+scale(Length)+
scale(Width_Mouth)+scale(Active_Farms)+scale(Temp)+scale(Sal)+
scale(Oxy)+Month_n+TimeSea+Class+Company+scale(Fallow)+
scale(Latitude),
ziformula=~scale(Depth)+scale(Distance_Mouth)+scale(Length)+
scale(Width_Mouth)+scale(Active_Farms)+scale(Temp)+scale(Sal)+
scale(Oxy)+Month_n+TimeSea+Class+Company+scale(Fallow)+
scale(Latitude),family=ziGamma(link="log"),data=SL_Farm)
m3.11<-glmmTMB(Lep~scale(Depth)+scale(Distance_Mouth)+scale(Length)+
scale(Width_Mouth)+scale(Active_Farms)+scale(Temp)+scale(Sal)+
scale(Oxy)+Month_n+Year+Class+Company+scale(Fallow)+
scale(Latitude),
ziformula=~scale(Depth)+scale(Distance_Mouth)+scale(Length)+
scale(Width_Mouth)+scale(Active_Farms)+scale(Temp)+scale(Sal)+
scale(Oxy)+Month_n+Year+Class+Company+scale(Fallow)+
scale(Latitude),family=ziGamma(link="log"),data=SL_Farm)
m3.12<-glmmTMB(Lep~scale(Depth)+scale(Distance_Mouth)+scale(Length)+
scale(Width_Mouth)+scale(Active_Farms)+scale(Temp)+scale(Sal)+
scale(Oxy)+Month_n+Year+TimeSea+Company+scale(Fallow)+
scale(Latitude),
ziformula=~scale(Depth)+scale(Distance_Mouth)+scale(Length)+
scale(Width_Mouth)+scale(Active_Farms)+scale(Temp)+scale(Sal)+
scale(Oxy)+Month_n+Year+TimeSea+Company+scale(Fallow)+
scale(Latitude),family=ziGamma(link="log"),data=SL_Farm)
m3.13<-glmmTMB(Lep~scale(Depth)+scale(Distance_Mouth)+scale(Length)+
scale(Width_Mouth)+scale(Active_Farms)+scale(Temp)+scale(Sal)+
scale(Oxy)+Month_n+Year+TimeSea+Class+scale(Fallow)+
scale(Latitude),
ziformula=~scale(Depth)+scale(Distance_Mouth)+scale(Length)+
scale(Width_Mouth)+scale(Active_Farms)+scale(Temp)+scale(Sal)+
scale(Oxy)+Month_n+Year+TimeSea+Class+scale(Fallow)+
scale(Latitude),family=ziGamma(link="log"),data=SL_Farm)
m3.14<-glmmTMB(Lep~scale(Depth)+scale(Distance_Mouth)+scale(Length)+
scale(Width_Mouth)+scale(Active_Farms)+scale(Temp)+scale(Sal)+
scale(Oxy)+Month_n+Year+TimeSea+Class+Company+
scale(Latitude),
ziformula=~scale(Depth)+scale(Distance_Mouth)+scale(Length)+
scale(Width_Mouth)+scale(Active_Farms)+scale(Temp)+scale(Sal)+
scale(Oxy)+Month_n+Year+TimeSea+Class+Company+
scale(Latitude),family=ziGamma(link="log"),data=SL_Farm)
m3.15<-glmmTMB(Lep~scale(Depth)+scale(Distance_Mouth)+scale(Length)+
scale(Width_Mouth)+scale(Active_Farms)+scale(Temp)+scale(Sal)+
scale(Oxy)+Month_n+Year+TimeSea+Class+Company+scale(Fallow),
ziformula=~scale(Depth)+scale(Distance_Mouth)+scale(Length)+
scale(Width_Mouth)+scale(Active_Farms)+scale(Temp)+scale(Sal)+

```

```
scale(Oxy)+Month_n+Year+TimeSea+Class+Company+scale(Fallow),
family=ziGamma(link="log"),data=SL_Farm)
```

```
AIC(m3.1,m3.01,m3.02,m3.03,m3.04,m3.05,m3.06,m3.07,
    m3.08,m3.09,m3.10,m3.11,m3.12,m3.13,m3.14,m3.15)
```

```
##      df      AIC
## m3.1  39 4308.564
## m3.01 37 4315.295
## m3.02 37 4336.555
## m3.03 37 4422.980
## m3.04 37 4305.694
## m3.05 37 4307.557
## m3.06 37 4314.669
## m3.07 37 4308.923
## m3.08 37 4360.802
## m3.09 37 4319.470
## m3.10 37 4346.165
## m3.11 33 5472.748
## m3.12 37 4329.858
## m3.13 35 4317.394
## m3.14 37 4304.979
## m3.15 37 4320.354
```

```
BIC(m3.1,m3.01,m3.02,m3.03,m3.04,m3.05,m3.06,m3.07,
    m3.08,m3.09,m3.10,m3.11,m3.12,m3.13,m3.14,m3.15)
```

```
##      df      BIC
## m3.1  39 4542.721
## m3.01 37 4537.444
## m3.02 37 4558.704
## m3.03 37 4645.129
## m3.04 37 4527.844
## m3.05 37 4529.706
## m3.06 37 4536.818
## m3.07 37 4531.072
## m3.08 37 4582.951
## m3.09 37 4541.619
## m3.10 37 4568.314
## m3.11 33 5670.882
## m3.12 37 4552.007
## m3.13 35 4527.535
## m3.14 37 4527.129
## m3.15 37 4542.503
```

Best fit was determined for the models m3.04, m3.05 and m3.14 without width of bays mouth, number of active farms, and fallow period. A new model was fitted excluding both of these predictors.

```
m3.2<-glmmTMB(Lep~scale(Depth)+scale(Distance_Mouth)+scale(Length)+
  scale(Temp)+scale(Sal)+scale(Oxy)+Month_n+Year+TimeSea+Class+
  Company+scale(Latitude),
  ziformula=~scale(Depth)+scale(Distance_Mouth)+scale(Length)+
```

```

scale(Temp)+scale(Sal)+scale(Oxy)+Month_n+Year+TimeSea+
Class+Company+scale(Latitude),family=ziGamma(link="log"),
data=SL_Farm)

AIC(m3.1,m3.04,m3.05,m3.14,m3.2)

```

```

##      df      AIC
## m3.1  39 4308.564
## m3.04 37 4305.694
## m3.05 37 4307.557
## m3.14 37 4304.979
## m3.2  33 4302.561

```

The final model was tested for overdispersion and the explained variance was determined.

```

# Overdispersion
E2 <- resid(m3.2, type = "pearson")
N <- nrow(SL_Farm)
p <- length(coef(m3.2))
sum(E2^2) / (N - p)

```

```
## [1] 2.288121
```

```

# Explained variance
model_performance(m3.2)

```

```

## # Indices of model performance
##
## AIC      |      AICc |      BIC | Nagelkerke's R2 |  RMSE | Sigma
## -----|-----|-----|-----|-----|-----
## 4302.561 | 4303.319 | 4500.694 |           0.110 | 1.543 | 1.252

```

The results of the final model were checked.

```

summary(m3.2)

## Family: Gamma ( log )
## Formula:      Lep ~ scale(Depth) + scale(Distance_Mouth) + scale(Length) +
##      scale(Temp) + scale(Sal) + scale(Oxy) + Month_n + Year +
##      TimeSea + Class + Company + scale(Latitude)
## Zero inflation:
## ~scale(Depth) + scale(Distance_Mouth) + scale(Length) + scale(Temp) +
##      scale(Sal) + scale(Oxy) + Month_n + Year + TimeSea + Class +
##      Company + scale(Latitude)
## Data: SL_Farm
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##    4302.6    4500.7   -2118.3    4236.6     2960
##
##
## Dispersion estimate for Gamma family (sigma^2): 1.57

```

```
##
## Conditional model:
##           Estimate Std. Error z value Pr(>|z|)
## (Intercept)      -1.806724   0.189919  -9.513  < 2e-16 ***
## scale(Depth)      -0.063483   0.035108  -1.808   0.0706 .
## scale(Distance_Mouth) 0.013323   0.040012   0.333   0.7392
## scale(Length)     -0.602540   0.057763 -10.431  < 2e-16 ***
## scale(Temp)        0.100098   0.039293   2.547   0.0109 *
## scale(Sal)         0.024360   0.032330   0.753   0.4512
## scale(Oxy)        -0.256663   0.033818  -7.589 3.21e-14 ***
## Month_n           0.285882   0.137139   2.085   0.0371 *
## Year              -0.474579   0.077836  -6.097 1.08e-09 ***
## TimeSea12          1.071139   0.111538   9.603  < 2e-16 ***
## TimeSea18          1.887786   0.108052  17.471  < 2e-16 ***
## TimeSea24          2.392032   0.123599  19.353  < 2e-16 ***
## ClassS1/2         -0.330259   0.067925  -4.862 1.16e-06 ***
## CompanyMowi        0.001863   0.137336   0.014   0.9892
## CompanyOthers      0.026225   0.134216   0.195   0.8451
## scale(Latitude)    0.208415   0.050095   4.160 3.18e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Zero-inflation model:
##           Estimate Std. Error z value Pr(>|z|)
## (Intercept)        0.16441    0.29547   0.556 0.577923
## scale(Depth)        0.14786    0.05592   2.644 0.008192 **
## scale(Distance_Mouth) 0.32128    0.05606   5.731 9.98e-09 ***
## scale(Length)       0.50530    0.09745   5.185 2.16e-07 ***
## scale(Temp)        -0.12859    0.06427  -2.001 0.045417 *
## scale(Sal)         0.14963    0.05729   2.612 0.009009 **
## scale(Oxy)         0.04775    0.04917   0.971 0.331491
## Month_n           0.75548    0.22858   3.305 0.000949 ***
## Year              0.36162    0.11863   3.048 0.002301 **
## TimeSea12         -1.71486    0.12480 -13.740 < 2e-16 ***
## TimeSea18         -3.17172    0.13957 -22.725 < 2e-16 ***
## TimeSea24         -3.44079    0.19280 -17.846 < 2e-16 ***
## ClassS1/2        -0.25195    0.11567  -2.178 0.029392 *
## CompanyMowi       0.19975    0.22831   0.875 0.381626
## CompanyOthers     0.70672    0.22120   3.195 0.001398 **
## scale(Latitude)   -0.01024    0.08423  -0.122 0.903233
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
car::Anova(m3.2)
```

```
## Analysis of Deviance Table (Type II Wald chisquare tests)
##
## Response: Lep
##           Chisq Df Pr(>Chisq)
## scale(Depth)      3.2697 1    0.07057 .
## scale(Distance_Mouth) 0.1109 1    0.73916
## scale(Length)    108.8096 1 < 2.2e-16 ***
## scale(Temp)       6.4896 1    0.01085 *
## scale(Sal)        0.5677 1    0.45116
```

```
## scale(Oxy)          57.5995  1  3.213e-14 ***
## Month_n            4.3456  1   0.03710 *
## Year               37.1758  1  1.079e-09 ***
## TimeSea           500.0924  3 < 2.2e-16 ***
## Class              23.6399  1  1.162e-06 ***
## Company            0.0886  2   0.95666
## scale(Latitude)    17.3090  1  3.177e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Visualization of results

The sea lice values for all levels of predictors and length of bay with the confidential interval were predicted.

```
newdata1<- data.frame(
  Depth= mean(scale(SL_Farm$Depth)),
  Distance_Mouth= mean(scale(SL_Farm$Distance_Mouth)),
  Length= rep(seq(from = min(SL_Farm$Length),
                      to = max(SL_Farm$Length),length.out = 100),72),
  Temp=mean(as.vector(SL_Farm$Temp)),
  Sal=mean(as.vector(SL_Farm$Sal)),
  Oxy=mean(as.vector(SL_Farm$Oxy)),
  Year= mean(as.vector(SL_Farm$Year)),
  Month_n=mean(as.vector(SL_Farm$Month_n)),
  TimeSea=factor(rep(rep(rep(1:4),3),2),each=100),levels=1:4,
                  labels=levels(SL_Farm$TimeSea)),
  Class= factor(rep(rep(1:2,36),each=100),levels=1:2,
                  labels=levels(as.factor(SL_Farm$Class))),
  Company=factor(rep(rep(rep(1:3,12),2),each=100),levels=1:3,
                  labels=levels(as.factor(SL_Farm$Company))),
  Latitude=mean(SL_Farm$Latitude))

newdata1 <- cbind(newdata1, predict(m3.2, newdata1, type = "link",
                                   se.fit=TRUE,re.form=NA))

newdata1 <- within(newdata1, {
  Abun<- exp(fit)
  LL <- exp(fit - 1.96 * se.fit)
  UL <- exp(fit + 1.96 * se.fit)
})
```

Mean values of sea lice year were calculated from predicted sea lice counts and the data was visualized.

```
A<-aggregate(newdata1$Abun,list(newdata1$Length),mean)
B<-aggregate(newdata1$LL,list(newdata1$Length),mean)
C<-aggregate(newdata1$UL,list(newdata1$Length),mean)
colnames(A)<-c('Length','Abun')
colnames(B)<-c('Length','LL')
colnames(C)<-c('Length','UL')
AA<-merge(A,B, by="Length")
BB<-merge(AA,C, by="Length")

P1<-ggplot(BB, aes(Length,Abun)) +
```

```

    geom_ribbon(aes(ymin=LL,ymax=UL),colour='grey80',fill= "grey80", alpha = 0.5)+
    geom_point()+
    geom_line(linewidth = 1.5)+
    labs(x = "Length of bay [km]", y = "")+
    theme_minimal(base_size = 12) +
    ylim(0,6)+
    theme(legend.position = "none")

```

The sea lice values for all levels of predictors and temperature with the confidential interval were predicted.

```

newdata1<- data.frame(
  Depth= mean(scale(SL_Farm$Depth)),
  Distance_Mouth= mean(scale(SL_Farm$Distance_Mouth)),
  Length=mean(scale(SL_Farm$Length)),
  Temp=rep(seq(from = min(SL_Farm$Temp),
                    to = max(SL_Farm$Temp),length.out = 100),72),
  Sal=mean(as.vector(SL_Farm$Sal)),
  Oxy=mean(as.vector(SL_Farm$Oxy)),
  Year= mean(as.vector(SL_Farm$Year)),
  Month_n=mean(as.vector(SL_Farm$Month_n)),
  TimeSea=factor(rep(rep(rep(1:4),3),2),each=100),levels=1:4,
                  labels=levels(SL_Farm$TimeSea)),
  Class= factor(rep(rep(1:2,36),each=100),levels=1:2,
                  labels=levels(as.factor(SL_Farm$Class))),
  Company=factor(rep(rep(rep(1:3,12),2),each=100),levels=1:3,
                  labels=levels(as.factor(SL_Farm$Company))),
  Latitude=mean(SL_Farm$Latitude))

newdata1 <- cbind(newdata1, predict(m3.2, newdata1, type = "link",
                                   se.fit=TRUE,re.form=NA))

newdata1 <- within(newdata1, {
  Abun<- exp(fit)
  LL <- exp(fit - 1.96 * se.fit)
  UL <- exp(fit + 1.96 * se.fit)
})

```

Mean values of sea lice year were calculated from predicted sea lice counts and the data was visualized.

```

A<-aggregate(newdata1$Abun,list(newdata1$Temp),mean)
B<-aggregate(newdata1$LL,list(newdata1$Temp),mean)
C<-aggregate(newdata1$UL,list(newdata1$Temp),mean)
colnames(A)<-c('Temp','Abun')
colnames(B)<-c('Temp','LL')
colnames(C)<-c('Temp','UL')
AA<-merge(A,B, by="Temp")
BB<-merge(AA,C, by="Temp")

P2<-ggplot(BB, aes(Temp,Abun)) +
  geom_ribbon(aes(ymin=LL,ymax=UL),colour='grey80',fill= "grey80", alpha = 0.5)+
  geom_point()+
  geom_line(linewidth = 2)+
  ylim(0,6)+

```

```
labs(x = "Temperature [\u00B0C]", y = "")+
theme_minimal(base_size = 12) +
theme(legend.position = "none",
      axis.text.y=element_blank())
```

The sea lice values for all levels of predictors and dissolved oxygen with the confidential interval were predicted.

```
newdata1<- data.frame(
  Depth= mean(scale(SL_Farm$Depth)),
  Distance_Mouth= mean(scale(SL_Farm$Distance_Mouth)),
  Length= mean(scale(SL_Farm$Length)),
  Temp=mean(as.vector(SL_Farm$Temp)),
  Sal=mean(as.vector(SL_Farm$Sal)),
  Oxy=rep(seq(from = min(SL_Farm$Oxy), to = max(SL_Farm$Oxy),length.out = 100),72),
  Year= mean(as.vector(SL_Farm$Year)),
  Month_n=mean(as.vector(SL_Farm$Month_n)),
  TimeSea=factor(rep(rep(rep(1:4),3),2),each=100),levels=1:4,
                 labels=levels(SL_Farm$TimeSea)),
  Class= factor(rep(rep(1:2,36),each=100),levels=1:2,
                 labels=levels(as.factor(SL_Farm$Class))),
  Company=factor(rep(rep(rep(1:3,12),2),each=100),levels=1:3,
                  labels=levels(as.factor(SL_Farm$Company))),
  Latitude=mean(SL_Farm$Latitude))

newdata1 <- cbind(newdata1, predict(m3.2, newdata1, type = "link",
                                   se.fit=TRUE,re.form=NA))

newdata1 <- within(newdata1, {
  Abun<- exp(fit)
  LL <- exp(fit - 1.96 * se.fit)
  UL <- exp(fit + 1.96 * se.fit)
})
```

Mean values of sea lice year were calculated from predicted sea lice counts and the data was visualized.

```
A<-aggregate(newdata1$Abun,list(newdata1$Oxy),mean)
B<-aggregate(newdata1$LL,list(newdata1$Oxy),mean)
C<-aggregate(newdata1$UL,list(newdata1$Oxy),mean)
colnames(A)<-c('Oxy','Abun')
colnames(B)<-c('Oxy','LL')
colnames(C)<-c('Oxy','UL')
AA<-merge(A,B, by="Oxy")
BB<-merge(AA,C, by="Oxy")

P3<-ggplot(BB, aes(Oxy,Abun)) +
  geom_ribbon(aes(ymin=LL,ymax=UL),colour='grey80',fill= "grey80", alpha = 0.5)+
  geom_point()+
  geom_line(linewidth = 1.5)+
  xlab("Dissolved oxygen [%]")+
  ylab('')+
  ylim(0,6)+
  theme_minimal(base_size = 12) +
```

```
theme(legend.position = "none",
      axis.text.y=element_blank())
```

All three graphs were plotted together.

```
grid.arrange(P1,P2,P3,ncol = 3,
             left =textGrob('Ovigerous female sea lice abundance \n [mean per farm]',
                           gp=gpar(fontsize=15),rot=90))

grid.text("a",x=unit(0.06, "npc"),y=unit(.95, "npc"),
          gp=gpar(fontsize=15, col="black"))
grid.text("b",x=unit(0.385, "npc"),y=unit(.95, "npc"),
          gp=gpar(fontsize=15, col="black"))
grid.text("c",x=unit(0.71, "npc"),y=unit(.95, "npc"),
          gp=gpar(fontsize=15, col="black"))
```

